

Sparse $\ell_2 - \ell_0$ image reconstruction

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Projet MORPHEME - UCA, CNRS, INRIA -

M2 MScDAI

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1. Introduction and examples
 2. ℓ_1 promotes sparsity
 3. Algorithms for $\ell_2 - \ell_1$ optimization
 4. Algorithms for $\ell_2 - \ell_0$ optimization
 5. Results on super-resolution Microscopy by Single Molecule Localization.
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1.Introduction

Many signal processing areas are concerned with

- ▶ Linear observation : $Ax = d$
 - ▶ d : observed data, vector in $\mathbb{R}^M$
 - ▶ x unknown data to be estimated in $\mathbb{R}^N$
 - ▶ A observation matrix, $M \times N$ matrix.

where we have **few observations** for a **large explicative unknown variables** x $M \ll N$

The system is undertermined, A is ill-conditioned, observations are noisy

- ▶ Least square solution $\hat{x} = \arg \min_{x \in \mathbb{R}^N} \|Ax - d\|_2^2$

$$(\|x\|_2^2 = \|x\|^2 = \sum_{i=1}^N x_i^2)$$

- ▶ Regularization: **sparse** signal hypothesis modeled by considering ℓ_1 -norm or ℓ_0 semi-norm constraints:

$$\|x\|_1 \leq K \text{ where } \|x\|_1 = \sum_{i=1}^N |x_i|$$

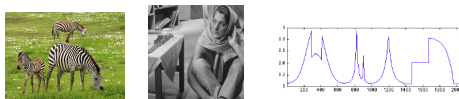
$$\|x\|_0 \leq K \text{ where } \|x\|_0 = \# \{x_i, i = 1, \dots, N : x_i \neq 0\}$$

NB: ℓ_0 -norm **is NOT** a norm as $\|\lambda x\|_0 = \|x\|_0 \neq \lambda \|x\|_0$.

1.0 Dictionary representation in image processing

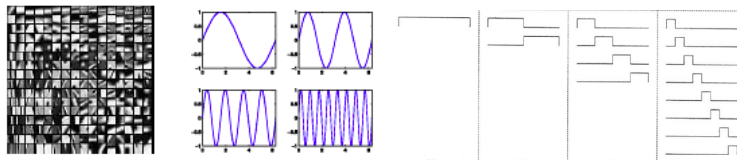
- Image are non-stationary, they exhibit smooth areas, oscillations, edges, textures,...

Let's $d \in \mathbb{R}^M$ be a patch of an image or a signal:



- Each part is represented by given waveforms which best match the image structure, for example Basis B_i as Haar, smooth wavelets, sine/cosine transform,...

Let's $A = [a_1, \dots, a_N] \in \mathbb{R}^{M \times N}$ be a set of basis vectors, or normalized vectors



1.0 Dictionary representation in image processing

- ▶ Such A is a redundant **dictionary** (succession of representative waveforms, possibly a succession of bases)
- ▶ The dictionary A is adapted to the signal d if d can be represented by a **few** number of vectors of the dictionary A , that is $d \approx Ax$ with x is a **sparse** vector, that is $\|x\|_0 \leq K$, where $K \ll N$.

$$\begin{bmatrix} d \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1 \begin{bmatrix} a_{11} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \end{bmatrix} + x_3 \begin{bmatrix} a_{13} \end{bmatrix} + \dots$$

1.1 Examples in Signal/image Processing

- ▶ signal is a sum of pulses, spikes, modeled by a sum of Dirac $\sum_{r=1}^K x_r \delta_{t_r}$.
- ▶ acquisition system, channel, is modeled as a linear system, e.g. convolution by a Gaussian function:
$$d(\cdot) = h * \sum_{r=1}^K x_r \delta_{t_r} = \sum_{r=1}^K x_r h(\cdot - t_r).$$

By assuming the Dirac locations t_r are on a regular grid indexed by $i = 1, \dots, N$

$$\begin{bmatrix} \vdots \\ d \\ \vdots \end{bmatrix} = \begin{bmatrix} \text{---} \end{bmatrix} \begin{bmatrix} x_1 & -t_1 \\ x_2 & -t_2 \\ x_3 & -t_3 \end{bmatrix} + \begin{bmatrix} \vdots \\ n \\ \vdots \end{bmatrix}$$

$d = A x + n$

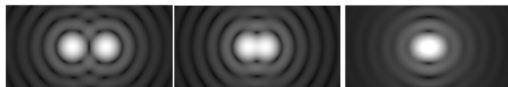
1D example: Channel estimation in communications, ...

2D example: Single Molecule Localization in super-resolution microscopy , ...

2D example in Super-resolution microscopy: SMLM (continued)

Conventional fluorescence microscopy limits

- ▶ physical diffraction limit of optical systems
- ▶ Airy patch = impulse response of the microscope (PSF: *Point Spread Function*)
- ▶ overlapping patches limit at $\approx 200\text{nm}$ the distance between two molecules to be resolved (Rayleigh limit)



Super-resolution by single molecule localization

- ▶ **Photo-activable molecules:** PALM *Photo Activated Localisation Microscopy* ([Betzig & al 06, Hess & al, 2006]) et STORM *STochastic Optical Reconstruction Microscopy* ([Rust & al, 2006])
- ▶ Sequentially activate and image a small random set of fluorescent molecules.

2D example in Super-resolution microscopy: SMLM (continued)

- ▶ activation
- ▶ imaging
- ▶ localization
- ▶ assembling



Figure: PALM microscopy principle. From Zeiss tutorials
[<http://zeiss-campus.magnet.fsu.edu/tutorials/index.html>]

2D example in Super-resolution microscopy: SMLM (continued)

- ▶ activation
- ▶ imaging
- ▶ localization
- ▶ assembling

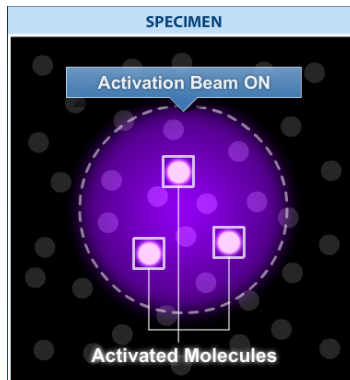


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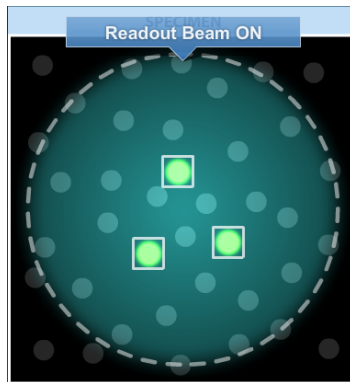


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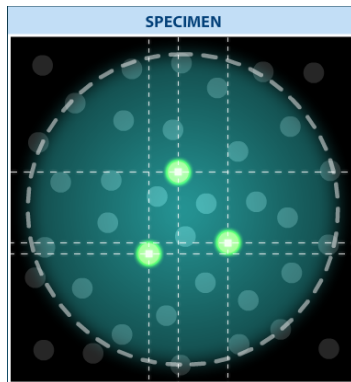


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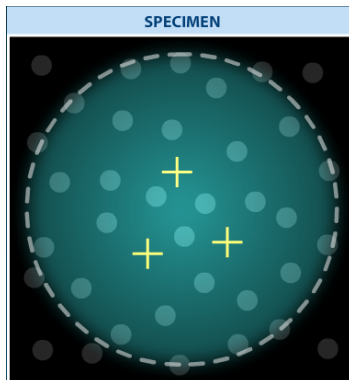


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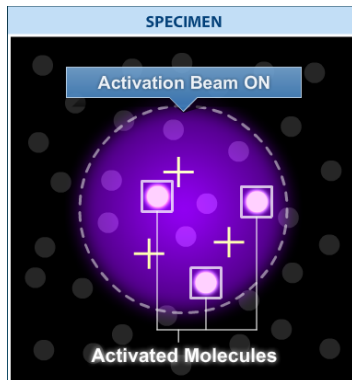


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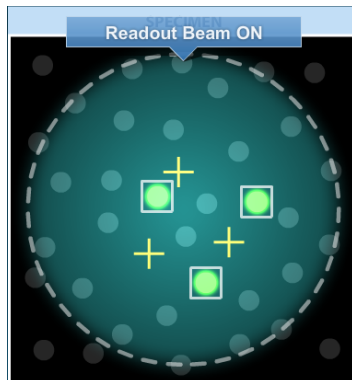


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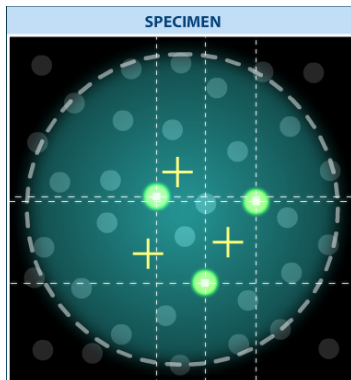


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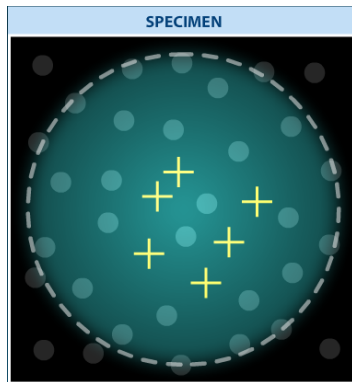
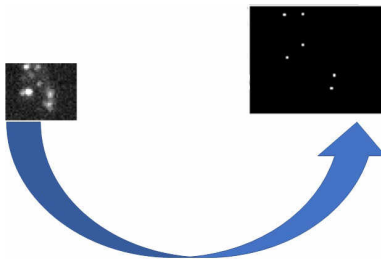


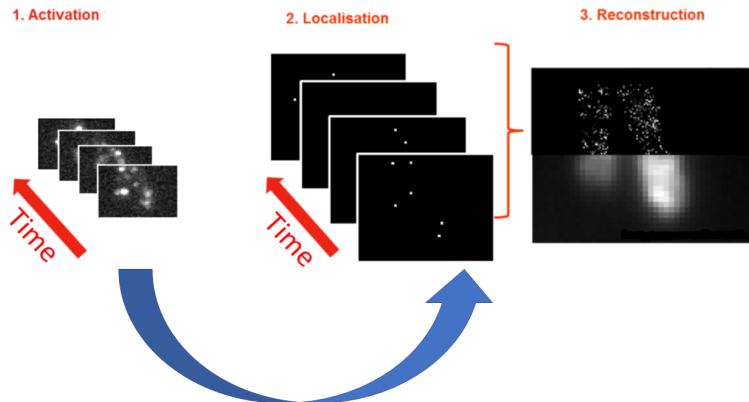
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2D example in Super-resolution microscopy: SMLM (continued)



$$\hat{\mathbf{x}} \in \arg \min_{\mathbf{x}} \frac{1}{2} \|\mathbf{Ax} - \mathbf{d}\|_2^2 + \lambda \|\mathbf{x}\|_0$$

2D example in Super-resolution microscopy: SMLM (continued)



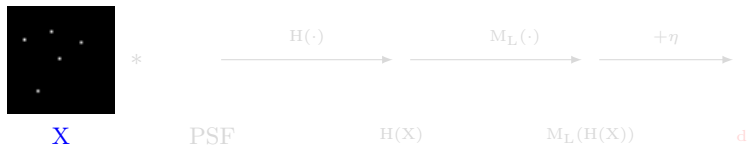
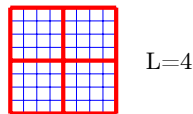
$$\hat{\mathbf{x}} \in \arg \min_{\mathbf{x}} \frac{1}{2} \|\mathbf{A}\mathbf{x} - \mathbf{d}\|_2^2 + \lambda \|\mathbf{x}\|_0$$

2D example in Super-resolution microscopy: SMLM (continued)

Image formation model PALM / STORM

$\mathbf{d} \in \mathbb{R}^{M \times M}$ one acquisition.

$\mathbf{X} \in \mathbb{R}^{ML \times ML}$ an image where each pixel of $\mathbf{d}$ is divided in $\mathbf{L} \times \mathbf{L}$ pixels.

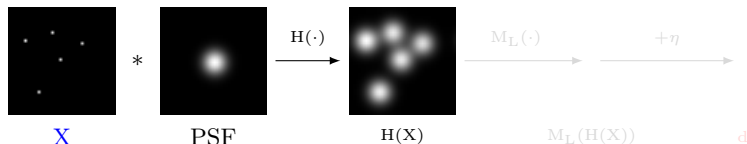
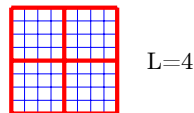


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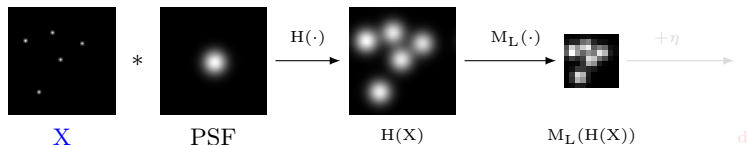
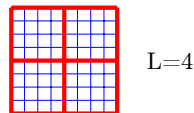


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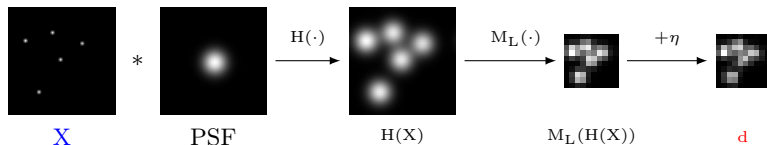
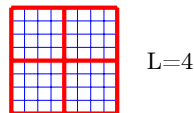


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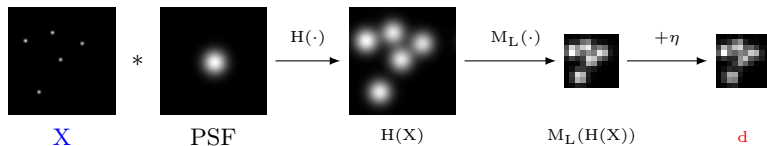
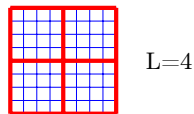


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Model

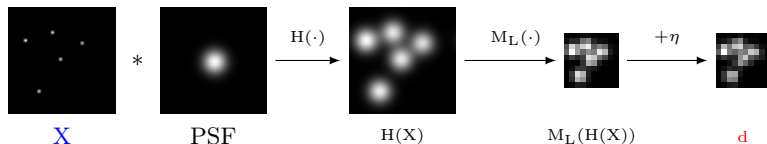
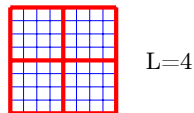
$$\mathbf{d} = M_L(H(\mathbf{X})) + \eta,$$

2D example in Super-resolution microscopy: SMLM (continued)

Image formation model PALM / STORM

$\mathbf{d} \in \mathbb{R}^{M \times M}$ one acquisition.

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Problem $\ell_2 - \ell_0$

$$\hat{\mathbf{X}} \in \arg \min_{\mathbf{X}} \frac{1}{2} \|\mathbf{d} - M_L(H(\mathbf{X}))\|_2^2 + \lambda \|\mathbf{X}\|_0$$

$\|\mathbf{X}\|_0 = \#\{X_i / X_i \neq 0\}$ is the number of non zero components of $\mathbf{X}$.

1.3 ℓ_2 - ℓ_0 optimization problems

Noisy problem: two constrained forms ($\epsilon > 0, K > 0$)

$$\hat{x} = \arg \min_{x \in \mathbb{R}^N} \frac{1}{2} \|Ax - d\|_2^2 \quad \text{subject to} \quad \|x\|_0 \leq K$$

$$\hat{x} = \arg \min_{x \in \mathbb{R}^N} \|x\|_0 \quad \text{subject to} \quad \|Ax - d\|_2^2 \leq \epsilon$$

Noisy problem : penalized form ($\lambda > 0$)

$$\hat{x} = \arg \min_{x \in \mathbb{R}^N} G_{\ell_0}(x) := \frac{1}{2} \|Ax - d\|_2^2 + \lambda \|x\|_0$$

$$A \in \mathbb{R}^{M \times N} \text{ with } M \ll N$$

- ▶ Non equivalent formulations
- ▶ Existence of an optimal solution and relationships between optimal solutions in [Nikolova 16]
- ▶ Intensive work in signal and image processing, and in statistics.
- ▶ **non-continuous**, **non-convex** and **NP-hard** optimization problem. [Natarajan 95] [Davis & al 97]. Roughly speaking, *a solution cannot be verified in polynomial time w.r.t the dimension of the problem*

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1. Iterative Hard Thresholding,
 2. Continuous relaxation,
 3. Greedy algorithms,
 4. Exact reformulation.
-

Penalized form

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x} \in \mathbb{R}^N} \frac{1}{2} \|A\mathbf{x} - d\|_2^2 + \lambda \|\mathbf{x}\|_0$$

- ▶ $\frac{1}{2} \|A\mathbf{x} - d\|_2^2$ is L -gradient Lipschitz ($L = \|A\|^2$)
- ▶ Proximal of $\|\cdot\|_0$ has explicit expression, this is the Hard Threshold

Iterative Hard Thresholding

(IHT): Forward-Backward Splitting (FBS) algorithm

$$\mathbf{x}^{k+1} = \text{prox}_{\gamma\lambda\|\cdot\|_0} \left(\mathbf{x}^k - \gamma A^t (A\mathbf{x}^k - d) \right)$$

$\gamma < \frac{1}{L}$ is the gradient step.

Computation of $\text{prox}_{\gamma\lambda\|\cdot\|_0}$:

$$\text{prox}_{\gamma\lambda\|\cdot\|_0}(y) = \arg \min_{x \in \mathbb{R}^N} \left\{ \frac{1}{2} \|x - y\|^2 + \gamma\lambda \|x\|_0 \right\}$$

$$\frac{1}{2} (x - y)^2 + \gamma\lambda \|x\|_0 = \sum_{i=1}^N (x_i - y_i)^2 + \gamma\lambda |x_i|_0$$

where $|u|_0 = 1$ if $u \neq 0$, 0 elsewhere.

Then it is sufficient to compute in 1D

$$\arg \min_{u \in \mathbb{R}} \{g(u) := \frac{1}{2}(u - y)^2 + \gamma\lambda |u|_0\}$$

IHT Algorithm (continued)

Computation of $\arg \min_{u \in \mathbb{R}} \{g(u) := \frac{1}{2}(u - y)^2 + \gamma\lambda|u|_0\}$



- ▶ if $u = 0$ then

$$g(0) = \frac{1}{2}(y)^2$$

- ▶ The minimum could be reached at $\hat{u} = 0$, the value is

$$g(\hat{u}) = \frac{1}{2}(y)^2$$

- ▶ if $u \neq 0$ then

$$g(u) = \frac{1}{2}(u - y)^2 + \lambda$$

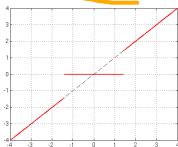
- ▶ The minimum is reached at $\hat{u} = y$ and the value is $g(\hat{u}) = \lambda$

if $|y| \leq \sqrt{2\lambda}$ then $\hat{u} = 0$

if $|y| \geq \sqrt{2\lambda}$ then $\hat{u} = y$

The solution is given by the Hard Threshold function

$$\hat{u} = \begin{cases} y & \text{if } |y| > \sqrt{2\lambda}, \\ 0 & \text{if } |y| \leq \sqrt{2\lambda}. \end{cases}$$



non convex

Find the solution of the optimal problem

$$\hat{x} = \arg \min_{x \in \mathbb{R}^N} \frac{1}{2} \|Ax - d\|_2^2 + \lambda \|x\|_0$$

by Forward Backward Splitting algorithm (Iterative Hard Thresholding)

$$x^{k+1} = \text{prox}_{\gamma\lambda\|\cdot\|_0} \left(x^k - \gamma A^t (Ax^k - d) \right)$$

- ▶ IHT algorithm converges to a critical point [Blumensath and Davies 08, Attouch et al 13].
- ▶ **Initialization** point is important, for example initialize with the solution with the ℓ_1 -norm problem:

$$\arg \min_{x \in \mathbb{R}^N} \left\{ \frac{1}{2} \|Ax - y\|^2 + \gamma\lambda \|x\|_1 \right\}. \text{ It is not guaranty that this}$$

solution is sparse.

4.3 ℓ_2 - ℓ_0 optimization by continuous relaxation

Continuous separable relaxation (convex and non-convex)

$$\frac{1}{2}\|Ax - d\|_2^2 + \lambda\|\mathbf{x}\|_0 \rightarrow \frac{1}{2}\|Ax - d\|_2^2 + \lambda\sum_{i \in \mathbb{I}_N} \phi(x_i)$$

Continuous approximation of the ℓ_0 -norm function:

- ▶ ℓ_1 -norm: Lasso [Tibshirani 96] ; Basic Pursuit [Chen et al 98] ; Compressed Sensing [Donoho 06, Candes et al 06]
- ▶ Adaptive Lasso [Zou 06] ;
- ▶ Nonnegative Garrote [Breiman 95] ;
- ▶ Exponential approximation [Mangasarian 96] ;
- ▶ Log-Sum Penalty [Candes et al 08] ;
- ▶ Smoothly Clipped Absolute Deviation (SCAD) [Fan and Li 01] ;
- ▶ Minimax Concave Penalty (MCP) [Zhang 10] ;
- ▶ ℓ_p -norms $0 < p < 1$ [Chartrand 07, Foucart and Lai 09] ;
- ▶ Smoothed ℓ_0 -norm Penalty (SL0) [Mohimani et al 09] ;

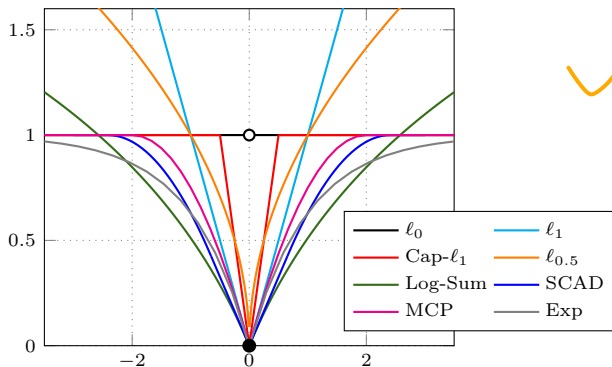
Are they *good* approximations?
Which one to use?

4.3 ℓ_2 - ℓ_0 optimization by continuous relaxation

Continuous separable relaxation (convex and non-convex)

$$\frac{1}{2}\|Ax - d\|_2^2 + \lambda\|x\|_0 \rightarrow \frac{1}{2}\|Ax - d\|_2^2 + \lambda\sum_{i \in \mathbb{I}_N} \phi(x_i)$$

Continuous approximation of the ℓ_0 -norm function:



Are they *good* approximations?
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4.3 ℓ_2 - ℓ_0 optimization by continuous relaxation

Example of continuous approximation functions of the ℓ_0 -norm:

- ▶ ℓ_1 -norm: $\phi(t) = |t|$
- ▶ Log-Sum Penalty [Candes et al 08] $\phi_{Log}(\theta; t) := \log(1 + |t|\theta)$, with $\theta \in \mathbb{R}_+^*$.
- ▶ Minimax Concave Penalty (MCP) [Zhang 10]
$$\phi_{MCP}(\gamma, \lambda; t) = \lambda \left(\frac{\gamma\lambda}{2} \mathbb{1}_{\{|t| > \gamma\lambda\}} + \left(|t| - \frac{t^2}{2\gamma\lambda} \right) \mathbb{1}_{\{|t| \leq \gamma\lambda\}} \right)$$

with $\mathbb{1}_{\{x \in C\}} = 1$ if $x \in C$ and 0 otherwise.

$$G_{\ell_0}(x) := \frac{1}{2} \|Ax - d\|_2^2 + \lambda \|x\|_0 \rightarrow \tilde{G}(x) := \frac{1}{2} \|Ax - d\|_2^2 + \sum_{i=1}^N \phi(x_i)$$

Definition of a *good* continuous approximation

- ▶ $G_{\ell_0}(x)$ and $\tilde{G}(x)$ have **same global** minimizers

$$\arg \min_{x \in \mathbb{R}^N} \tilde{G}(x) = \arg \min_{x \in \mathbb{R}^N} G_{\ell_0}(x) \quad (\text{P1})$$

- ▶ $\tilde{G}(x)$ has **less local** minimizers than $G_{\ell_0}(x)$

$$\hat{x} \text{ minimiseur de } \tilde{G} \implies \hat{x} \text{ minimiseur de } G_{\ell_0} \quad (\text{P2})$$

ℓ_2 - ℓ_0 optimization by continuous relaxation

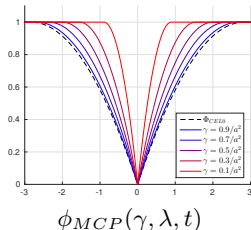
ϕ depends on $\|a_i\|$ and λ when applied on x_i :

$$\tilde{G}(x) := \frac{1}{2} \|Ax - d\|_2^2 + \sum_{i \in \mathbb{I}_N} \phi(\|a_i\|, \lambda, x_i)$$

The one which removes the most of local minimizers is $\phi_{MCP}(\frac{1}{\|a_i\|}, \lambda, t)$ that we call ϕ_{CELO} :

$$\phi_{CELO}(\|a_i\|, \lambda, x) = \lambda - \frac{\|a_i\|^2}{2} \left(|x| - \frac{\sqrt{2\lambda}}{\|a_i\|} \right)^2 \mathbb{1}_{\{|x| \leq \frac{\sqrt{2\lambda}}{\|a_i\|}\}}$$

where $\mathbb{1}_{\{x \in D\}} = 1$ if $x \in D$; 0 otherwise.



The $\ell_2 - \ell_0$ and ℓ_2 -CELO functionals :

$$G_{\ell_0}(x) := \frac{1}{2} \|Ax - d\|^2 + \lambda \|x\|_0$$

$$G_{\text{CELO}}(x) = \frac{1}{2} \|Ax - d\|^2 + \sum_{i \in \mathbb{I}_N} \phi_{\text{CELO}}(\|a_i\|, \lambda, x_i)$$

where $\phi_{\text{CELO}}(\|a_i\|, \lambda, x) = \lambda - \frac{\|a_i\|^2}{2} \left(|x| - \frac{\sqrt{2\lambda}}{\|a_i\|} \right)^2 \mathbb{1}_{\{|x| \leq \frac{\sqrt{2\lambda}}{\|a_i\|}} \}$

Properties of $G_{\text{CELO}}(x)$

- ▶ **Limit inf** of the functions satisfying (P1) and (P2) ; the one which removes the most of local minimizers
- ▶ **Continuity**
- ▶ **Non convex** in the general case (for any A)
- ▶ but **convexity** with respect to each **component**

Nonsmooth nonconvex algorithms

The **continuity of G_{CELO}** allows to use recent *nonsmooth nonconvex* algorithms to minimize (indirectly) G_{ℓ_0} ,

- ▶ *Difference of Convex* (DC) functions programming [[Gasso et al 09](#)]
- ▶ *Majorization-Minimization* (MM) algorithms (*e.g.* Iteratively Reweighted ℓ_1 (IRL1) [[Ochs et al 2015](#)])
- ▶ *Forward-Backward splitting* (GIST [[Gong et al 13](#)], [[Attouch et al 13](#)])

ℓ_2 - ℓ_0 optimization by continuous relaxation

Forward-Backward Splitting Algorithm

$$\mathbf{x}^{k+1} \in \text{prox}_{\gamma \Phi_{\text{CELO}}(\cdot)} \left(\mathbf{x}^k - \gamma^k A^T (A \mathbf{x}^k - d) \right),$$

where $0 < \gamma < \frac{1}{\|A\|^2}$ and

$$\text{prox}_{\gamma \phi_{\text{CELO}}(a, \lambda; \cdot)}(u) = \begin{cases} \text{sign}(u) \min \left(|u|, (|u| - \sqrt{2\lambda}\gamma a)_+ / (1 - a^2\gamma) \right) & \text{if } \underline{a^2\gamma} < 1 \\ u \mathbf{1}_{\{|u| > \sqrt{2\gamma\lambda}\}} + \{0, u\} \mathbf{1}_{\{|u| = \sqrt{2\gamma\lambda}\}} & \text{if } a^2\gamma \geq 1 \end{cases}$$

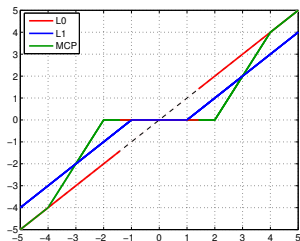


Figure: Proximal operators. Red: ℓ_0 , Blue: ℓ_1 , Green: Φ_{CELO} (depends on $a = \|a_i\|$ at component $u = x_i$).



Forward-Backward Splitting Algorithm

$$\mathbf{x}^{k+1} \in \text{prox}_{\gamma \Phi_{\text{CELO}}(\cdot)} \left(\mathbf{x}^k - \gamma^k A^T (A \mathbf{x}^k - d) \right),$$

where $0 < \gamma < \frac{1}{\|A\|^2}$ and

$$\text{prox}_{\gamma \phi_{\text{CELO}}(a, \lambda; \cdot)}(u) = \begin{cases} \text{sign}(u) \min \left(|u|, (|u| - \sqrt{2\lambda}\gamma a)_+ / (1 - a^2\gamma) \right) & \text{if } a^2\gamma < 1 \\ u \mathbb{1}_{\{|u| > \sqrt{2\gamma\lambda}\}} + \{0, u\} \mathbb{1}_{\{|u| = \sqrt{2\gamma\lambda}\}} & \text{if } a^2\gamma \geq 1 \end{cases}$$

- Convergence to a critical point under Kurdyka-Lojaseiwicz (KL) property [Attouch et al 13].
- Accelerated algorithm in the non convex case [Li Lin 15]

5.1 Results, ISBI challenge 2013, simulated dataset

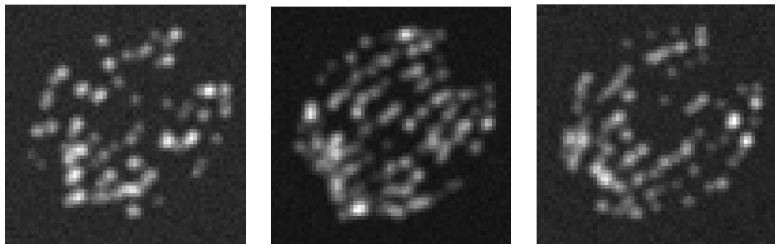


Figure: Simulated images (among the 361 simulated high density images for this sample). Data from IEEE ISBI Challenge 2013.

<http://bigwww.epfl.ch/smlm/datasets/index.html>

8 simulated tubes of 30nm diameter

Camera of 64×64 pixels of size 100nm.

Gaussian PSF, FWHM = 258.21 nm (full width at half maximum)

80932 molecules activated on 361 frames.

5.1 Results, ISBI challenge 2013, simulated dataset

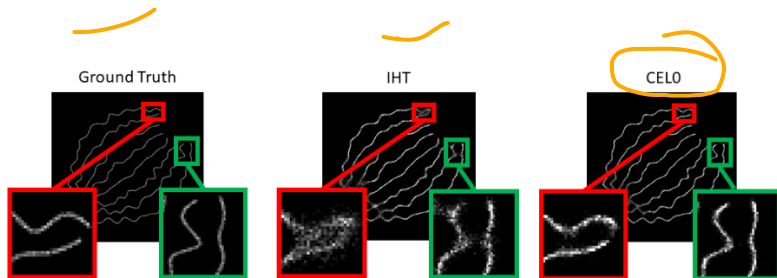


Figure: Reconstruction from simulated data set, reduction ratio $L = 4$.

Greedy algorithms

Greedy algorithms, *Matching Pursuit (MP)* [Mallat et al 93], *Orthogonal MP* [Pati et al 93], *Orthogonal Least Squares (OLS)* [Chen et al 89], *Bayesian OMP* [Herzet et al 10], *Single Best Replacement* [Soussen et al 11] and further variants.

Matching Pursuit:

d is the signal we want to represent with the a limited number $K \ll N$ of waveforms or atoms of dictionary A , one atom is one column of A , *i.e.* $A_{:,i} = a_i$, $i = 1, ..N$.

$$\begin{bmatrix} d \end{bmatrix} = \begin{bmatrix} | & | & | & | \\ a_{11} & a_{12} & a_{13} & \dots \\ | & | & | & | \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \end{bmatrix} = x_1 \begin{bmatrix} | \\ a_{11} \\ | \end{bmatrix} + x_2 \begin{bmatrix} | \\ a_{12} \\ | \end{bmatrix} + x_3 \begin{bmatrix} | \\ a_{13} \\ | \end{bmatrix} + \dots$$

For that we have to solve

$$\hat{x} = \arg \min_{x \in \mathbb{R}^N} \|Ax - d\|_2^2 \quad \text{subject to} \quad \|x\|_0 \leq K.$$

$$(\text{ or } \hat{x} = \arg \min_{x \in \mathbb{R}^N} \|x\|_0 \quad \text{subject to} \quad \|Ax - d\|_2^2 \leq \epsilon)$$

Greedy algorithms (continued)

Matching Pursuit principle

It is assumed without loss of generality that A has unit norm columns, $\|A_{:,i}\| = \|a_i\| = 1$.

The **first component** $i^1 \in \{1, \dots, N\}$ will be such that the **correlation** between d and atom i is maximum: $i^1 = \arg \max_{j \in \{1, \dots, N\}} |\langle a_j, d \rangle|$.

Then the **optimal solution** is $x^1 = (0, 0, \dots, \langle a_{i^1}, d \rangle, 0, \dots, 0)$, where the non null component is at index i^1 , which is written as $x^1 = \langle a_{i^1}, d \rangle \cdot e_{i^1}$, $e_i \in \mathbb{R}^N$, $i \in \{1, \dots, N\}$ is the canonical basis in $\mathbb{R}^N$.

The criterion is $\|A \cdot x^1 - d\|^2 = \|d\|^2 - (\langle a_{i^1}, d \rangle)^2$.

The **residual** is $r = d - A \cdot x^1 = d - \langle a_{i^1}, d \rangle a_{i^1}$, and the process is repeated.

Matching Pursuit Algorithm

Input: A (with unit norm column), d , K .

Initialize: $r^0 = d, \sigma^0 = \emptyset, (x^0 = 0)$.

Repeat, while $\#\sigma^k \leq K$: (or while $\|r^k\| > \epsilon$)

$$\begin{aligned} i^k &= \arg \max_{j \in \{1, \dots, N\}} |\langle r^k, a_j \rangle| \\ \sigma^{k+1} &= \sigma^k \cup \{i^k\} \\ r^{k+1} &= r^k - \langle r^k, a_{i^k} \rangle a_{i^k} \end{aligned}$$

(1)

σ^k is the support of the current solution x^k , that is the indexes of the non-zero components. $\#\sigma^k$ is the cardinal of σ^k . The initial value of $\#\sigma^0$ is 0 and it increases by 1 at each iteration.

The optimal solution at current iteration is $x^{k+1} = x^k + \langle r^k, a_{i^k} \rangle a_{i^k}$.

- ▶ The residual $\|r^k\|$ converges exponentially to 0 [Mallat et al 93].
- ▶ Sub-optimal solution: retro-project the residual onto $\text{Span}\{(a_i)_{i \in \sigma^k}\}$ reduce the approximation error ($\|A x^K - d\|^2$).

Greedy algorithms (continued)

Orthogonal Matching Pursuit [Pati et al 93, Tropp 04]: at each iteration, optimally estimate the intensities with the current support of the solution fixed, by $\mathbf{x}^{k+1} = \arg \min_{\{\mathbf{x} / \sigma_{\mathbf{x}} \subset \sigma^{k+1}\}} \|\mathbf{A}\mathbf{x} - \mathbf{d}\|^2$.

Orthogonal Matching Pursuit (OMP) Algorithm Input: $\mathbf{A}$ (with

unit norm column), $\mathbf{d}$, K .

i = intensity

Initialize: $\mathbf{r}^0 = \mathbf{d}$, $\sigma^0 = \emptyset$

Repeat, while $\#\sigma^k \leq K$:

$$i^k = \arg \max_{j \notin \sigma^k} |\langle \mathbf{r}^k, \mathbf{a}_j \rangle|$$

$$\sigma^{k+1} = \sigma^k \cup \{i^k\}$$

$$\mathbf{x}^{k+1} = \arg \min_{\{\mathbf{x} / \sigma_{\mathbf{x}} \subset \sigma^{k+1}\}} \|\mathbf{A}\mathbf{x} - \mathbf{d}\|^2$$

$$\mathbf{r}^{k+1} = \mathbf{d} - \mathbf{A}\mathbf{x}^{k+1}$$

- ▶ Convergence in N iterations at most (at each iteration a **new** component is selected),
- ▶ Exact sparse recovery results (under conditions on $\mathbf{A}$) [Tropp 04].

Greedy algorithms (continued)

Further algorithms:

At each iteration, several strategies for one component to be

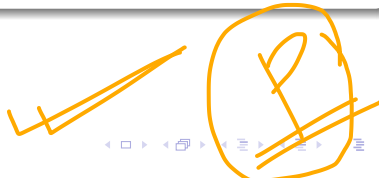
- ▶ added,
- ▶ removed,
- ▶ replaced.

Orthogonal Least Squares (OLS) [Chen et al 89], *Bayesian OMP* [Herzet et al 10], *Single Best Replacement* [Soussen et al 11] and further variants [Jain & al 11, Soussen et al 15]...

The more complex is the strategy, the best is the solution and the longest is the computing time.

Exact reformulation

- ▶ *Class of continuous nonconvex penalties* → asymptotic connections with the ℓ_2 - ℓ_0 criteria [Chouzenoux et al 13]
- ▶ *Reformulation using Difference of Convex functions* → asymptotic or local minimizer results [Le Thi et al 14, Le Thi et al 15]
- ▶ *Equivalence of ℓ_0 - and ℓ_p -norm ($0 < p \leq 1$) minimization under linear equalities or inequalities (e.g. exact reconstruction problem)* [Fung and Mangasarian 11]
- ▶ *Reformulation and optimization through Mixed-Integer Programs (MIPs)* → global optimum for problems of reasonable size (a few hundred variables) [Bourguignon et al 15]
- ▶ **Exact reformulation** ([Bi et al 14, Yuan & Ghanem 16, Liu et al 18], ...)



Exact reformulation of ℓ_0 : Penalized reformulation

Lemma 1 [Liu et al 18, Yuan & Ghanem 16]

$$\|x\|_0 = \min_{-1 \leq u \leq 1} \|u\|_1 \text{ s.t. } \|x\|_1 = \langle u, x \rangle$$

Exact reformulation for the $\ell_2 - \ell_0$ penalized problem

Initial problem:

$$\min_x \frac{1}{2} \|Ax - d\|_2^2 + \lambda \|x\|_0$$

Penalized reformulation:

$$\min_{x,u} G_\rho(x, u) := \frac{1}{2} \|Ax - d\|_2^2 + \iota_{\{-1 \leq \cdot \leq 1\}}(u) + \lambda \|u\|_1 + \rho (\|x\|_1 - \langle x, u \rangle)$$

with $\iota_{\{x \in D\}}(x) = 0$ if $x \in D$, $+\infty$ otherwise.

Theorem [Bechensteen, et al.]

If $\rho > \sigma_{\max}(A) \|d\|_2$, and A is of full rank. Then:

1. If (x_ρ, u_ρ) is a local (respectively global) minimizer of G_ρ , then x_ρ is a local (respectively global) minimizer of the initial problem.
2. If $\hat{x}$ is a global minimizer of the initial problem, then $(\hat{x}, \hat{u})$ is a global minimizer of G_ρ with $\hat{u}$ associated with Lemma 1.

Exact reformulation of ℓ_0 : Constrained reformulation

Lemma 1 [Liu et al 18, Yuan & Ghanem 16]

$$\|x\|_0 = \min_{-1 \leq u \leq 1} \|u\|_1 \text{ s.t. } \|x\|_1 = \langle u, x \rangle$$

Exact reformulation for the $\ell_2 - \ell_0$ constrained problem

Initial problem:

$$\min_x \frac{1}{2} \|Ax - d\|_2^2 + \iota_{\{\|\cdot\|_0 \leq K\}}(x)$$

Constrained reformulation:

$$\min_{x,u} G_\rho(x, u) := \frac{1}{2} \|Ax - d\|^2 + \iota_{\{\cdot \geq 0\}}(x) + \iota_{\{-1 \leq \cdot \leq 1\}}(u) + \iota_{\{\|\cdot\|_1 \leq K\}}(u) + \rho(\|x\|_1 - \langle x, u \rangle)$$

Theorem [Bechensteen, et al.]

If $\rho > \sigma_{\max}(A)\|d\|_2$, and A is of full rank. Then:

1. If (x_ρ, u_ρ) is a local (respectively global) minimizer of G_ρ , then x_ρ is a local (respectively global) minimizer of the initial problem.
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Why minimize the constrained or penalized reformulation instead of their initial formulation?

Constrained reformulation:

$$\min_{\mathbf{x}, \mathbf{u}} \frac{1}{2} \|A\mathbf{x} - d\|^2 + \iota_{\{\cdot \geq 0\}}(\mathbf{x}) + \iota_{\{-1 \leq \cdot \leq 1\}}(\mathbf{u}) + \iota_{\{\|\cdot\|_1 \leq K\}}(\mathbf{u}) + \rho(\|\mathbf{x}\|_1 - \langle \mathbf{x}, \mathbf{u} \rangle)$$

Penalized reformulation:

$$\min_{\mathbf{x}, \mathbf{u}} \frac{1}{2} \|A\mathbf{x} - d\|^2 + \iota_{\{\cdot \geq 0\}}(\mathbf{x}) + \iota_{\{-1 \leq \cdot \leq 1\}}(\mathbf{u}) + \lambda \|\mathbf{u}\|_1 + \rho(\|\mathbf{x}\|_1 - \langle \mathbf{x}, \mathbf{u} \rangle)$$

- ▶ Biconvex
- ▶ Non-convexity linked to the coupling term $\langle \mathbf{x}, \mathbf{u} \rangle$
- ▶ Minimizing the reformulation is equivalent to minimize the initial problem regarding local and global minimizers

Exact reformulation of ℓ_0 : Algorithm

We add a positivity constraint on x and we finally define

$$G_\rho(x, u) = \frac{1}{2} \|Ax - d\|^2 + \iota_{\{x \geq 0\}}(x) + \rho \|x\|_1 + \iota_{\{\|\cdot\|_1 \leq K\}}(u) + \iota_{\{-1 \leq \cdot \leq 1\}}(u) - \rho < x, u$$

The global optimization scheme is (continuation method)

Initialize: $\rho^0 > 0, n = 0$

Repeat: Solve the problem G_{ρ^n} :

$$\{x^{n+1}, u^{n+1}\} = \arg \min_{x, u} G_{\rho^n}(x, u)$$

Update: $\rho^{n+1} = \alpha \rho^n, \alpha > 1$

Until: $\rho^{n+1} > \sigma_{\max}(A) \|d\|_2$

Exact reformulation of ℓ_0 : Algorithm

$$G_{\rho^n}(x, u) = \frac{1}{2} \|Ax - d\|^2 + \iota_{\{\cdot \geq 0\}}(x) + \rho^n \|x\|_1 + \iota_{\{\|\cdot\|_1 \leq K\}}(u) + \iota_{\{-1 \leq \cdot \leq 1\}}(u) - \rho^n < \infty$$

At fixed ρ^n we apply the Proximal Alternate Minimization (PAM) algorithm [Attouch & al 10]

Initialize: $u^0 = 0 \in \mathbb{R}^M$

Repeat: $\arg \min_x G_{\rho^n}(x, u^n)$ using alternate minimizations

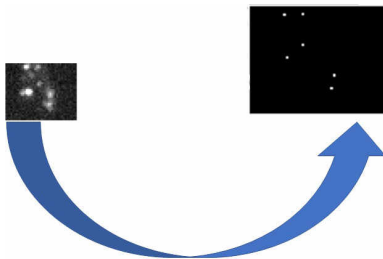
$$\begin{aligned} \blacktriangleright \{x^{n+1}\} &= \arg \min_x G_{\rho^n}(x, u^n) + \frac{1}{2c^n} \|x - x^n\|^2 \\ &\rightarrow \text{FISTA Algorithm [Beck et al 09]} \end{aligned}$$

$$\begin{aligned} \blacktriangleright \{u^{n+1}\} &= \arg \min_u G_{\rho^n}(x^{n+1}, u) + \frac{1}{2d^n} \|u - u^n\|^2 \\ &\rightarrow \text{Algorithm [Stefanov, 2004]} \end{aligned}$$

Until: convergence

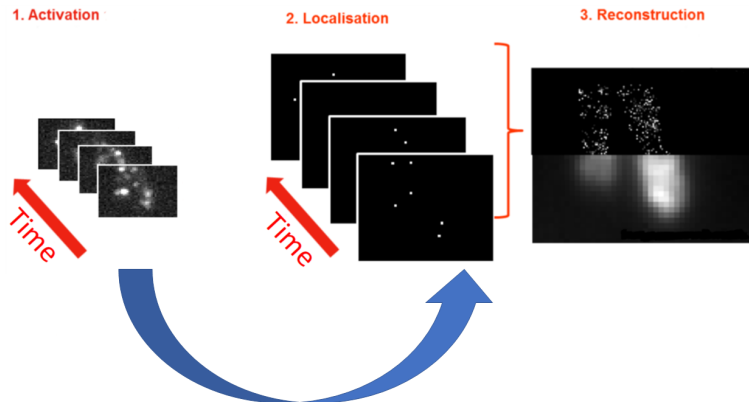
Convergence of the algorithm towards a critical point of G_{ρ^n} for c^n and d^n such that $0 < r_- < c^n, d^n < r_+$ and under KL condition on G_{ρ^n} and assuming that x_n and u_n are bounded [Attouch & al 10].

Results: Single-Molecule Localization Microscopy



$$\hat{\mathbf{x}} \in \arg \min_{\mathbf{x}} \frac{1}{2} \|\mathbf{Ax} - \mathbf{d}\|_2^2 + \iota_{\{\cdot \geq 0\}}(\mathbf{x}) + R(\mathbf{x})$$

Results: Single-Molecule Localization Microscopy



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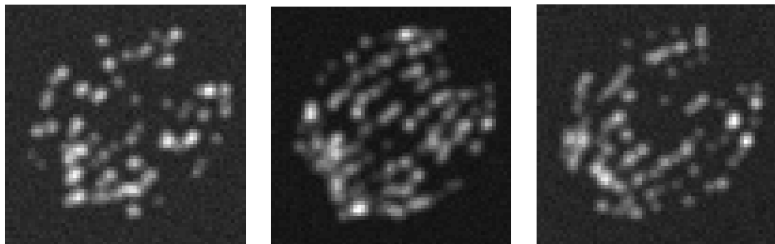


Figure: Simulated images (among the 361 simulated high density images for this sample). Data from IEEE ISBI Challenge 2013.

<http://bigwww.epfl.ch/smlm/datasets/index.html>

8 simulated tubes of 30nm diameter

Camera of 64×64 pixels of size 100nm.

Gaussian PSF, FWHM = 258.21 nm (full width at half maximum)

80932 molecules activated on 361 frames.

Results, ISBI challenge 2013, simulated dataset

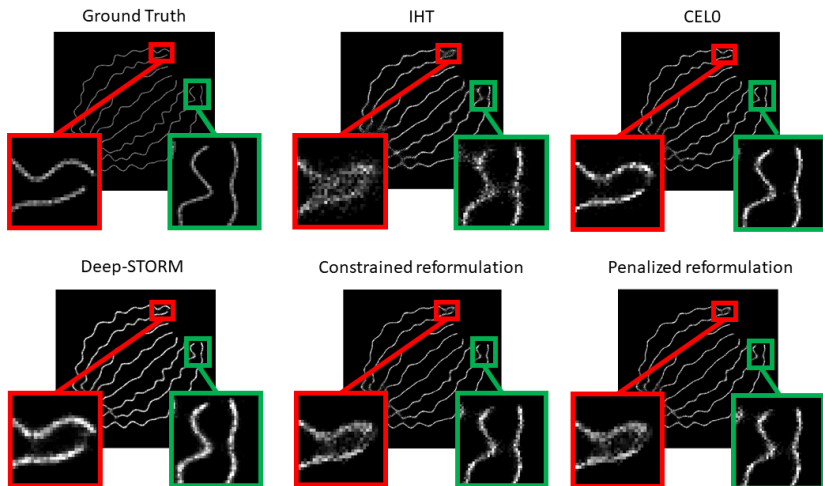
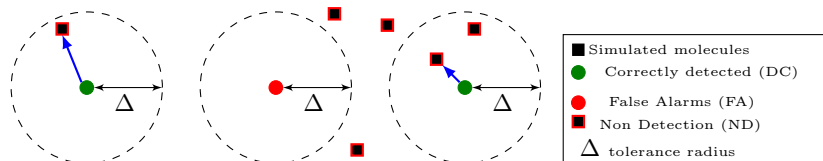


Figure: Reconstruction from simulated data set, reduction ratio $L = 4$.

Jaccard index calculus



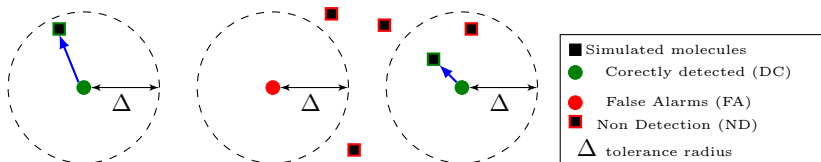
$$\text{Jaccard index} = \frac{\text{DC}}{\text{DC} + \text{FA} + \text{ND}}$$

Jaccard index results

	Jaccard index (%)			
Method - Tolerance (nm)	50	100	150	200
IHT	20.1	35.9	40.4	41.3
CEL0	29.3	41.3	42.4	42.6
Constrained reformulation	25.2	40.0	43.2	43.9
Penalized reformulation	25.0	39.3	42.2	42.8
Deep-STORM	×	×	×	×

Table: The Jaccard index obtained and the tolerance

Jaccard index calculus



$$\text{Jaccard index} = \frac{\text{DC}}{\text{DC} + \text{FA} + \text{ND}}$$

Jaccard index results

	Jaccard index (%)			
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Table: The jaccard index obtained and the tolerance

Results, ISBI challenge 2013, Real dataset

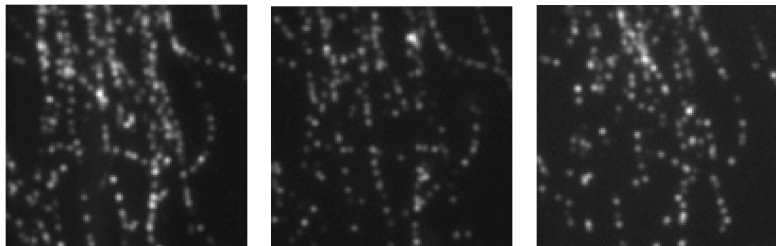


Figure: Real images (among the 500 real high density images for this sample).
Data from IEEE ISBI Challenge 2013.

<http://bigwww.epfl.ch/smlm/datasets/index.html>

Camera of 128×128 pixels of size 100nm.

Gaussian PSF, FWHM = 358.1 nm (full width at half maximum)

Results, ISBI challenge 2013, Real dataset

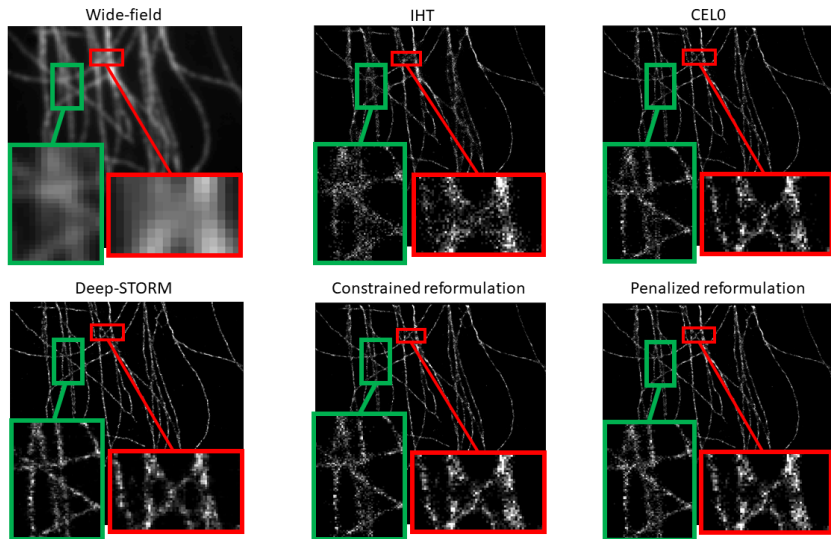


Figure: Reconstruction from the real data set, reduction ratio $L = 4$.

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